

Assignment 2

1. $f(x) = kx(1-x)$, $0 \leq x \leq 1$

(i) Find k , (ii) $P(x \leq b) = P(x \geq b) \rightarrow b = ?$, (iii) Find CDF

(i) PDF $\rightarrow \int_{-\infty}^{\infty} f(x) dx = 1$

$$\int_0^1 kx(1-x) dx = 1 \rightarrow k \int_0^1 x - x^2 dx = 1$$

$$= k \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = 1 \rightarrow k \left[\frac{1}{2} - \frac{1}{3} \right] = 1 \rightarrow \boxed{k = 6}$$

(ii) $P(x \leq b) = P(x \geq b)$

$$6 \int_0^b x - x^2 dx = 1 - 6 \int_0^b x - x^2 dx$$

$$= 6 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^b = 1 - 6 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^b$$

$$= 6 \left[\frac{b^2}{2} - \frac{b^3}{3} \right] = 1 - 6 \left[\frac{b^2}{2} - \frac{b^3}{3} \right]$$

$$= 12 \left[\frac{b^2}{2} - \frac{b^3}{3} \right] = 1 \rightarrow 12 \left[\frac{3b^2 - 2b^3}{6} \right] = 1 \rightarrow 2(3b^2 - 2b^3) = 1$$

$$= 2b^3 - 3b^2 + \frac{1}{2} = 0 \rightarrow 0 \leq b \leq 1 \rightarrow \boxed{b = \frac{1}{2} = 0.5}$$

(iii) CDF = ?

$$F_X(x) = \int_{-\infty}^x f_X(t) dx$$

$$= \int_0^x 6t(1-t) dt = 6 \left[\frac{t^2}{2} - \frac{t^3}{3} \right]_0^x$$

$$= 6 \left[\frac{x^2}{2} - \frac{x^3}{3} \right] = 3x^2 - 2x^3$$

$$F_X(x) = 3x^2 - 2x^3 \quad \text{for } x \cdot 0 \leq x \leq 1$$

2. $F_X(x) = 1 - e^{-x^2/b}$; $b \geq 0$, $x > 0$

PDF = ?

$$\frac{d}{dx} F_X(x) = 0 - e^{-x^2/b} \cdot \frac{-2x}{b}$$

$$f_X(x) = \frac{2x}{b} \cdot e^{-x^2/b}$$

6. $f_X(x) = k(1+x)$, $2 \leq x \leq 5$

$$P(X < 4) = k \int_2^5 (1+x) dx = 1 \rightarrow k \left[x + \frac{x^2}{2} \right]_2^5 = 1$$

$$k \left[\frac{5}{1} + \frac{25}{2} - 2 - \frac{4}{2} \right] = 1 \rightarrow k \left[3 + \frac{21}{2} \right] = 1 \rightarrow k = \frac{2}{27}$$

$$k \left[\frac{6+21}{2} \right] = 1 \rightarrow k = \frac{2}{27}$$

$$P(X < 4) = \frac{2}{27} \int_2^4 (1+x) dx \rightarrow \frac{1}{8} \left[x + \frac{x^2}{2} \right]_2^4$$

$$= \frac{2}{27} \left[4 + \frac{16}{2} - 2 - \frac{4}{2} \right] = \frac{2}{27} [2 + 8]$$

$$= \boxed{\frac{16}{27}}$$

7. $f_X(x) = \begin{cases} k(1-x^2), & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$

(i) $k = ?$, (ii) $P(0.1 < x < 0.2)$, (iii) $P(x > 0.5)$

(i) $k = ?$

$$k \int_0^1 (1-x^2) dx = 1 \rightarrow k \left[x - \frac{x^3}{3} \right]_0^1 = 1$$

$$= k \left[1 - \frac{1}{3} \right] = 1 \rightarrow k \left[\frac{2}{3} \right] = 1 \rightarrow \boxed{k = \frac{3}{2}}$$

(ii) $\frac{3}{2} \int_{0.1}^{0.2} (1-x^2) dx \rightarrow \frac{3}{2} \left[x - \frac{x^3}{3} \right]_{0.1}^{0.2}$

$$= \frac{3}{2} \left[0.2 - \frac{(0.2)^3}{3} - 0.1 + \frac{(0.1)^3}{3} \right] = \frac{3}{2} \left[0.1 - \frac{7}{3000} \right]$$

$$= \frac{293}{3000} = \boxed{0.146}$$

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$$(iii) P(x > 0.5) = \frac{3}{2} \int_{0.5}^1 1 - x^2 dx$$

$$= \frac{3}{2} \left[x - \frac{x^3}{3} \right]_{0.5}^1 = \frac{3}{2} \left[1 - \frac{1}{3} - 0.5 + \frac{(0.5)^3}{3} \right]$$

$$= \frac{3}{2} \left[\frac{5}{24} \right] = \frac{5}{16} = \boxed{0.3125}$$

$$3. f_x(x) = \begin{cases} \frac{1}{x_0} e^{-x/x_0}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

$$(i) P(x > x_0) = \frac{1}{x_0} \int_{x_0}^{\infty} e^{-x/x_0} dx$$

$$= \frac{1}{x_0} \left[e^{-x/x_0} \cdot -x_0 \right]_{x_0}^{\infty} \rightarrow \frac{1}{x_0} \left[-e^{-x_0/x_0} - (-x_0) \right]$$

$$= e^{-x_0/x_0} = e^{-1} = \boxed{0.367}$$

$$(ii) P(x < x_0) = \frac{1}{x_0} \int_0^{x_0} e^{-x/x_0} dx$$

$$= \frac{1}{x_0} \left[e^{-x/x_0} \cdot -x_0 \right]_0^{x_0} \rightarrow \frac{1}{x_0} \left[e^{-1} \cdot -x_0 - 1 \cdot -x_0 \right]$$

$$= \frac{1}{x_0} \left[-x_0 \cdot e^{-1} + x_0 \right] = 1 - e^{-1} = 1 - 0.367 = \boxed{0.632}$$

3. Exponential $\rightarrow$ CDF $\rightarrow 1 - e^{-\lambda x}$

$$E(X) = \frac{1}{\lambda} \rightarrow \lambda = \frac{1}{x_0}$$

$$(i) P(X > x_0) = 1 - P(X \leq x_0) \\ = 1 - (1 - e^{-x/x_0})$$

5. $f_X(X) = \begin{cases} \frac{3}{4} (2x - x^2), & 0 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$

$$(i) P(X \leq 1) = \frac{3}{4} \int_0^1 (2x - x^2) dx = \frac{3}{4} \left[\frac{2x^2}{2} - \frac{x^3}{3} \right]_0^1 \\ = \frac{3}{4} \left[1 - \frac{1}{3} \right] = \frac{3}{4} \left[\frac{2}{3} \right] = \frac{1}{2} = \boxed{0.5}$$

$$(ii) P(X > 1.5) = \frac{3}{4} \int_{1.5}^2 (2x - x^2) dx = \frac{3}{4} \left[\frac{2x^2}{2} - \frac{x^3}{3} \right]_{1.5}^2 \\ = \frac{3}{4} \left[4 - \frac{8}{3} - \frac{9}{4} + \frac{9}{8} \right] = \frac{5}{32} = \boxed{0.156}$$

$$(iii) P(0.5 \leq X \leq 1.5) = \frac{3}{4} \int_{0.5}^{1.5} (2x - x^2) dx = \frac{3}{4} \left[\frac{2x^2}{2} - \frac{x^3}{3} \right]_{0.5}^{1.5} \\ = \frac{3}{4} \left[\frac{(1.5)^2}{2} - \frac{(1.5)^3}{3} - \left(\frac{(0.5)^2}{2} - \frac{(0.5)^3}{3} \right) \right] \\ = \frac{11}{16} = \boxed{0.6875}$$

$$4. \quad x(t) = \begin{cases} 3 \times 10^{-9} t^2 (100 - t)^2, & 0 \leq t \leq 100 \\ 0, & \text{elsewhere} \end{cases}$$

$$(i) \quad P(60 < x < 70) = 3 \times 10^{-9} \int_{60}^{70} t^2 \cdot (10000 + t^2 - 200t) dt$$

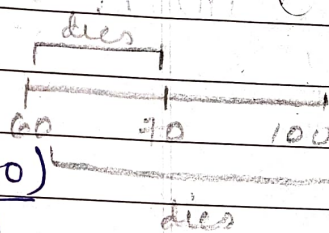
$$= 3 \times 10^{-9} \int_{60}^{70} (10000t^2 + t^4 - 200t^3) dt$$

$$= 3 \times 10^{-9} \left[10000 \cdot \frac{t^3}{3} + \frac{t^5}{5} - 200 \cdot \frac{t^4}{4} \right]_{60}^{70}$$

$$= 3 \times 10^{-9} \left[10000 \cdot \frac{(70)^3}{3} + \frac{(70)^5}{5} - 200 \cdot \frac{(70)^4}{4} - 10000 \cdot \frac{(60)^3}{3} - \frac{(60)^5}{5} + 200 \cdot \frac{(60)^4}{4} \right]$$

$$= 3 \times 10^{-9} [278973333.3 - 227520000]$$

$$= 3 \times 10^{-9} \times 51453333.3 = \boxed{0.154}$$



$$(ii) \quad P(60 < x < 70 | x \geq 60) = \frac{P(60 < x < 70)}{P(x > 60)}$$

$$P(x > 60) = 3 \times 10^{-9} \int_{60}^{100} (10000t^2 + t^4 - 200t^3) dt$$

$$= 3 \times 10^{-9} \left[10000 \cdot \frac{(100)^3}{3} + \frac{(100)^5}{5} - 200 \cdot \frac{(100)^4}{4} - 10000 \cdot \frac{(60)^3}{3} - \frac{(60)^5}{5} + 200 \cdot \frac{(60)^4}{4} \right]$$

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$$= 3 \times 10^{-9} [333333333.3 - 227520000] = 0.317$$

$$P(60 < x < 70) = 0.154 \rightarrow 0.154 / 0.317 = \boxed{0.4851}$$